

Role	Focus	Common Tools
33. Data Analyst	Reporting & interpre ^{te} interpreting existing data	Excel, SQL, PowerBI
Data Scientist	Building predictive models & statistical analysis	Python, R, Jupyter
Data Engineer	Building pipeline & infrastructure for data flow	SQL, Databricks, Spark
AI Engineer	Building & developing AI/ML applications	Python, TensorFlow, APIs

34. Business Intelligence is the practice of collecting and analyzing business data to support informed decision-making. PowerBI and Tableau are popular tools that turn raw data into interactive dashboards helping businesses spot trends & act on real-time insights.

35. LLMs and Generative AI is what I found interesting as I normally use Claude for building formulas and dashboards for work.

36. Neural Networks, I went to youtube for better understanding

37. It has made my day job much easier and faster. I normally use AI to build excel dashboard layouts and also to unpack formulas. Learning SQL has made it much easier for me to build ~~good~~ google studio dashboards there's a lot of coding happening there.

29. Data Privacy is the practice of protecting personal information from unauthorized access or misuse.

GDPR is a European Union law that sets strict rules on how personal data is collected & used applying to all EU citizens' data regardless of where the company processing is located.

30. Data Ethics involves the moral principles guiding responsible collection, use & sharing of data

eg of unethical practice include selling customer data without consent, using biased AI hiring tools that discriminate & surveilling employees without disclosure, each problematic because it violates trust, fairness or legal rights.

31. Data governance is the framework of policies & processes that ensure data is accurate, secure and used properly across an organization. Key components include data quality standards, clear ownership / accountability, security controls & compliance policies

32. Bias in AI occurs when a model produces unfair or skewed outcomes because its training data reflects historical inequalities or lacks diversity.
eg facial recognition system performing worse on darker skin tones due to underrepresented training data, which can lead to misidentification & unfair treatment.

27. A Jupyter Notebook is an interactive coding environment that lets users write and run code in small sections alongside text, visuals & output. it's most commonly used with Python & is popular among analysts and data scientists as it makes exploring data, testing code & documenting findings easy in one place.

28. Python is a high level, readable programming language widely used in data analytics & AI due to its simple syntax & huge ecosystem of libraries. Common libraries include Pandas (data manipulation), NumPy (numerical computing), Matplotlib (data visualization) & Scikit-learn (machine learning).

23. Prompt Engineering is the skill of crafting clear, specific instructions to get useful responses from AI ~~tools~~ tools like ChatGPT or Claude

eg "Write about sales" is vague whereas "Write a 3 sentence summary of Q2 sales trends for a retail client" gives a clear direction & produces better ~~res~~ results.

24. A MAP function applies a given operation to every item in a list or array and returns a new list of results.

In Python, `list(map(lambda x: x * 2, [1, 2, 3]))` doubles each number, returning `[2, 4, 6]`

25. Cloud computing is the delivery of computing services (storage, processing, software) over the internet instead of on local hardware. IaaS provide raw infrastructure like servers (AWS EC2), PaaS provides a platform to build applications without managing infrastructure (Google App Engine) SaaS delivers ready-to-use software over the internet (Microsoft 365)

26. A version control system tracks changes to files over time, allowing multiple people to collaborate without overwriting each other's work.

Git is the most widely used system as it's fast & flexible.

Commit saves a snapshot of changes, while a branch is a separate line of development that can be merged back later.

19. API is a set of rules that lets 2 softwares systems communicate with each other, much like a waiter taking an order from a customer to the kitchen & bringing back the food. APIs are important because they let different applications share data & functionality without needing to know each other's internal code.

20. An API key is a unique code used to identify and authenticate who is making a request to an API, controlling access & usage. It should be stored securely.

eg in environment variables, never in public code. because if exposed it may be misused.

21. REST API: REST = Representational State Transfer an architectural ~~sys~~ style for designing web APIs using standard HTTP methods. GET retrieves data POST creates new data, PUT updates existing data and DELETE removes data.

22. JSON is a lightweight, text-based data format used to store & exchange structured data between systems, especially in APIs. It's popular because it's easy for both humans to read and machines to parse.
eg `{ "name": "Thandeka", "role": "Supervisor" }`

- ChatGPT used for conversational assistance & content generation
- Claude used for reasoning, writing & analysis tasks.

16. Generative AI refers to models that create new content, such as text, images or audio by learning patterns from existing data and generating similar outputs.

- ChatGPT → text generation
- DALL-E → image generation
- GitHub & Copilot → code generation

17. A training dataset is the data used to teach a machine learning model to recognize patterns & make predictions.

Quality is critical because if the data is biased or incorrect the model will learn & repeat those flaws, leading to unfair or inaccurate real-world decisions.

18. Computer Vision is a field of AI that trains machines to interpret & understand visual info from images or videos, typically by breaking images into pixel data & using neural networks to detect patterns like edges and shapes.

- facial recognition for phones security & automated quality inspection of factory production lines.

11. AI is the simulation of human-like reasoning & decision-making by computer system. Narrow AI is designed for one specific task, like Siri answering questions while General AI would match full human cognitive ability across any task, something that doesn't exist yet outside theory.

12. ML is a subset of AI where systems learn pattern from data rather than being explicitly programmed with rules.

Supervised learning uses labeled data to predict outcomes (predicting sales from past data) unsupervised learning finds hidden patterns in unlabeled data (customer segmentation) reinforcement learning learns through trial and error using rewards and penalties (a game-playing AI)

13. Deep learning is a subset of machine learning that uses multi-layered neural networks to automatically learn complex patterns from large amounts of data. Neural networks work by passing data through interconnected layers of neurons each adjusting weights during training to improve accuracy over time.

14. NLP is a branch of AI focused on enabling computers to understand & generate human language. (voice assistants, like Siri, spam email filters & auto-suggest autocorrect on phones).

15. LLM is a type of AI trained on massive amounts of text data to understand & generate human like language, differing from standard ML models by its scale & general-purpose language ability rather than being built for one narrow task.

9. ETL = Extract, Transform and Load

↳ pulling data from source system, cleaning & reshaping it into a usable format and placing it into a target system like a warehouse.

It's important in data analytics because it ensures data from different sources is consistent, accurate & ready for reliable reporting and analysis

10. CSV is a simple text format good for small, straightforward tabular data, while JSON handles semi-structured, nested data well and is widely used in APIs. Parquet & Avro are optimized for big data processing, with Parquet being column base - fast for analytics queries and Avro being row based - better for streaming and serialization.

- Semi-structured data has some organizational tags but no fixed schema

eg a JSON file

- unstructured data has no predefined format

eg photos & videos.

each data type requires different tools & approaches to store & analyze.

6. Data modeling is the process of designing how data is structured, related and stored within a system before its built. It's important because good model prevents redundancy & errors & makes data easier to query.

common types include conceptual models (high-level business view) and relational models (tables with defined relationships eg star schema).

7. Data mining is the process of discovering patterns, trends & relationships in large datasets using statistical & computational techniques.

eg include retailers analyzing purchase history to recommend products and banks mining transactions data to be detecting fraud.

8. A Database Management System is a software that lets users create, store, manage & retrieve data in database.

MySQL and PostgreSQL are popular open-source relational DBMS good for web applications, Oracle Database is used for large enterprise system needing high reliable and MongoDB is a NoSQL DBMS suited for flexible unstructured data like user profiles or logs.

Research Assignment 1

1. An organized collection of data store electronically for easy access, manage & updating.
 - bank's customer account
 - retailer's inventory management system.
2. A large central repository that stores structured, historical data from multiple sources, the data is used for reporting and analysis. Companies use one to combine data from different systems. (sales, marketing, finance) into a single source for business intelligence & decision-making
3. Data lake stores raw data in its native format (structured, semi-structured or unstructured) without requiring it to be organized, unlike warehouse which needs cleaned structured data. Business prefer a data lake when having large volumes of varied data (log, images, social media) & want flexibility for future analysis such as machine learning
4. Data mart is a smaller, focused subset of a data warehouse built for a specific department or business function such as sales or marketing. It relates to the warehouse as a slice of it, typically used by business users who need quick access to relevant data without navigating the whole warehouse.
5. Structured data fits neatly into rows and columns, eg excel sales spreadsheet